

Group 36 Progress Report: Recurrent Neural Network-Based Weather Prediction

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1 Introduction

Our challenge is to construct a model that will predict immediate short-term weather patterns around Toronto, given a large sample of past weather information. There has been a lot of recent development with AI-based weather models, and there is no shortage of large, open-source weather datasets to train models with, so we figured this would be an interesting project to tackle. Because of this boom in AI weather models, many newer ones have been performing significantly better than traditional physics-based models, in both accuracy and speed [6]. Our approach to solving this problem is with a recurrent neural network, that can make use of the sequential structure of the data in order to leverage patterns in past weather conditions, allowing us to predict future short-term trends.

Many aspects of weather prediction are heavily time-based so we will try to capture both with our recurrent network. A good example of this is that when close to a large lake, we can predict that the chance of rain will increase after a period of high temperatures due to Lake-Effect precipitation [5]. Since data-driven models can capture patterns like this without needing the background reasoning of why or how it happens, it follows that they are overtaking physics based-models with recent developments [6], so we will compare our algorithm to physics-based models in order to properly judge its efficacy.

2 Related Work

One project we discovered involved generating accurate predictive data for several weather metrics [4]. This project was a neural network-based predictive algorithm, with several iterative versions of this model. The project experimented with comparing feed-forward neural networks to recurrent networks, eventually implementing gated recurrent unit to help with the vanishing gradient problem.

The algorithms each took sequential weather data and compiled it into a model to predict future trends of those data points, in order to make clean, more consistent synthetic data for other models to draw on.

Another project we looked into was a recurrent neural network weather prediction task that would classify images based on non-exclusive weather states [8]. They used self-annotated data, and their model would return a list of probabilities that each state was occurring. This algorithm initially was based on a single variable classifier, but the paper quickly concludes that a non-exclusive multi-class algorithm is the most efficient and accurate way to label outputs. This makes sense, as often a lot of weather conditions overlap, so reducing to just ‘wind’ or ‘rain’ is reductive, and can cause loss of information.

3 Dataset

Our dataset is a large sample of public domain weather data taken from the Government of Canada’s website through a provided Bash script. The dataset consists of measurements taken roughly every 3 hours, ranging from June 11th, 2013, to September 26th, 2025. The exact time between measurements is highly irregular, resulting in inconsistent amounts of labels on a day-to-day basis. Additionally, the table is full of null values, and many empty or extraneous columns, including two that were completely unlabelled.

Due to the frequency and time range of recordings, our dataset has 107,000 data points, which means we can afford to drop invalid or incomplete points.

3.1 Labels

The table is provided as a .csv file, with built-in labels for each point, classifying what weather conditions were currently unfolding at the given time.

These labels are grouped together and treat different magnitudes of the same weather as different prefix labels, for example “Mostly Cloudy” and “Partially Cloudy” are treated as separate cases. We want to combine these umbrella terms by adding a discrete numeric method of indicating the scale of each weather event. This will allow us to add similar such labels back into the output during post-processing.

3.2 Preprocessing

For our preprocessing, we removed missing or redundant values from the dataset. Since we only requested data from a station outside of Toronto’s Pearson International Airport, which means that all the location and station data columns are identical, so we can remove those. There was a large amount of mostly empty flag columns and the humidex was missing most of its values, we removed all of those features.

For the date and time information, we want to parse it manually to work with our recurrent neural network. We added in a delta time column, recording the time in hours since the last measurement, and parsed the day and month integer values into sinusoidal functions.

Lastly, we take our labels and split them into a list of each characteristic, then we group the labels into label classes, and convert the prefix labels into real valued magnitudes representing the scale of the labeled weather event. Finally we export the processed data into a separate file for later splitting into training and test sets.

4 Features

The features of our processed dataset include:

- Date Information (Day, Month and Year)
- Time (LST)
- Wind Speed (km/h)
- Wind Direction (10s of degrees)
- Air Pressure (kPa)
- Visibility (km)
- Relative Humidity (
- Dew Point Temperature ($^{\circ}C$)
- Precipitation Amount (mm)

Our input values contain all of this information, as we have filtered out any columns missing any values. These features were manually determined by human inspection of the dataset.

We engineered the date and time features to have more readable numeric values to be more compati-

ble with our recurrent neural network. We added in a delta time column, recording the time in hours since the last measurement, and parsed the day and month integer values into sinusoidal functions.

5 Implementation

Describe your model and implementation here. Refer to item 4. This may take around a page.

6 Results and Evaluation

How are you evaluating your model? What results do you have so far? What are your baselines? Refer to item 5. This may take around 0.5 pages.

7 Feedback and Plans

Write about your plans for the remainder of the project. This should include a discussion of the feedback you received from your TA, and how you plan to improve your approach. Reflect on your implementation and areas for improvement. Refer to item 6. This may be around 0.5 pages.

8 Template Notes

You can remove this section or comment it out, as it only contains instructions for how to use this template. You may use subsections in your document as you find appropriate.

8.1 Tables and figures

See Table 1 for an example of a table and its caption. See Figure ?? for an example of a figure and its caption.

8.2 Citations

Table 1 shows the syntax supported by the style files. We encourage you to use the natbib styles. You can use the command `\citet` (cite in text) to get “author (year)” citations, like this citation to a paper by Gusfield [3]. You can use the command `\citep` (cite in parentheses) to get “(author, year)” citations [3]. You can use the command `\citealp` (alternative cite without parentheses) to get “author, year” citations, which is useful for using citations within parentheses (e.g. 3).

8.3 References

Many websites where you can find academic papers also allow you to export a bib file for citation or bib formatted entry. Copy this into the `custom.bib` and you will be able to cite the paper in the $\text{\LaTeX}$. You can remove the example entries.

Output	natbib command	ACL only command
[3]	\citep	
3	\citealp	
Gusfield [3]	\citet	
[1997]	\citeyearpar	
Gusfield’s (1997)		\citeposs

Table 1: Citation commands supported by the style file.

8.4 Equations

An example equation is shown below:

$$A = \pi r^2 \quad (1)$$

Labels for equation numbers, sections, subsections, figures and tables are all defined with the `\label{label}` command and cross references to them are made with the `\ref{label}` command. This an example cross-reference to Equation 1. You can also write equations inline, like this: $A = \pi r^2$.

Team Contributions

Write in this section a few sentences describing the contributions of each team member. What did each member work on? Refer to item 7.

References

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- [4] Jung Min Han, Yu Qian Ang, Ali Malkawi, and Holly W. Samuelson. 2021. [Using recurrent neural networks for localized weather prediction with combined use of public airport data and on-site measurements](#). *Building and Environment*, 192:107601.
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- [7] Mohammad Sadegh Rasooli and Joel R. Tetreault. 2015. [Yara parser: A fast and accurate dependency parser](#). *Computing Research Repository*, arXiv:1503.06733. Version 2.
- [8] Bin Zhao, Xuelong Li, Xiaoqiang Lu, and Zhigang Wang. 2019. [A cnn-rnn architecture for multi-label weather recognition](#). *Preprint*, arXiv:1904.10709.